IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-499

1. Topic (12 words max)

Real-Time Safety Monitoring System for Public Transport in Sri Lanka

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In Sri Lanka, public transportation serves as a primary mode of travel for millions, yet passenger safety remains a persistent concern. Issues such as driver fatigue, traffic rule violations, and vehicle overloading contribute significantly to accidents and dissatisfaction among commuters. These challenges are further compounded by limited technological integration in the monitoring and enforcement of safety standards within the public transport sector.

The current infrastructure relies heavily on manual oversight, which is neither scalable nor effective in ensuring compliance with safety regulations. For example, bus drivers often operate long hours without adequate rest, increasing the risk of fatigue-related incidents. Additionally, there is limited data collection on real-time traffic violations or occupancy rates, making it difficult for authorities to act proactively or optimize transport operations.

The research problem centers around the lack of a unified, automated system that can provide real-time monitoring of key safety indicators. Addressing this requires a robust technological intervention capable of integrating multiple streams of sensor data—such as driver behavior, speed, passenger count, and geolocation—into a centralized platform for real-time decision-making.

To tackle this issue, this project proposes the design and implementation of **SmartBus Guardian**, an IoT-based safety monitoring system developed in collaboration with the **Centre of Excellence for AI**. The system will use embedded vision and sensing technologies to monitor driver alertness, detect rule violations, prevent overloading, and provide live tracking for passengers and authorities. This research aims to not only improve operational safety but also generate actionable insights for policy-makers and transport authorities to strengthen regulatory frameworks and infrastructure planning.

References

- Dissanayake, D., & Morikawa, T. (2010). Mode choice and air pollution impacts of public transport improvement in Colombo, Sri Lanka. *Journal of the Eastern Asia Society for Transportation Studies*, *8*, 1055–1070.
- Perera, L. H. M. R., & Dissanayake, D. (2022). Safety Concerns in Urban Public Transportation in Sri Lanka. *Transportation Research Forum*, *58*(3), 244–258.
- World Health Organization. (2018). *Global status report on road safety 2018*. WHO Press. https://www.who.int/publications/i/item/9789241565684
- Jayasinghe, A., & Perera, H. N. (2021). Developing Intelligent Transport Systems for Emerging Economies: A Case Study from Sri Lanka. *International Journal of Intelligent Transportation Systems Research*, *19*(2), 267–277.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

This project proposes **SmartBus Guardian**, a modular, AI-driven IoT solution designed to enhance safety in Sri Lanka's public transport sector. Developed in collaboration with the **Centre of Excellence for AI**, the system integrates four major subsystems:

1. **Driver Monitoring**: Utilizes an eye-tracking camera and seatbelt sensor to detect signs of drowsiness or distraction. Real-time alerts are triggered if unsafe conditions are identified.

2. **Rule Violation Detection**: It uses GPS, speed sensors, and gyroscope sensor to detect traffic violations such as speeding and traffic violations. It notifies the driver of up to 3 traffic violations within a certain distance. If it continues, a message is sent to the nearest traffic police officer's phone with the bus's location and other details.

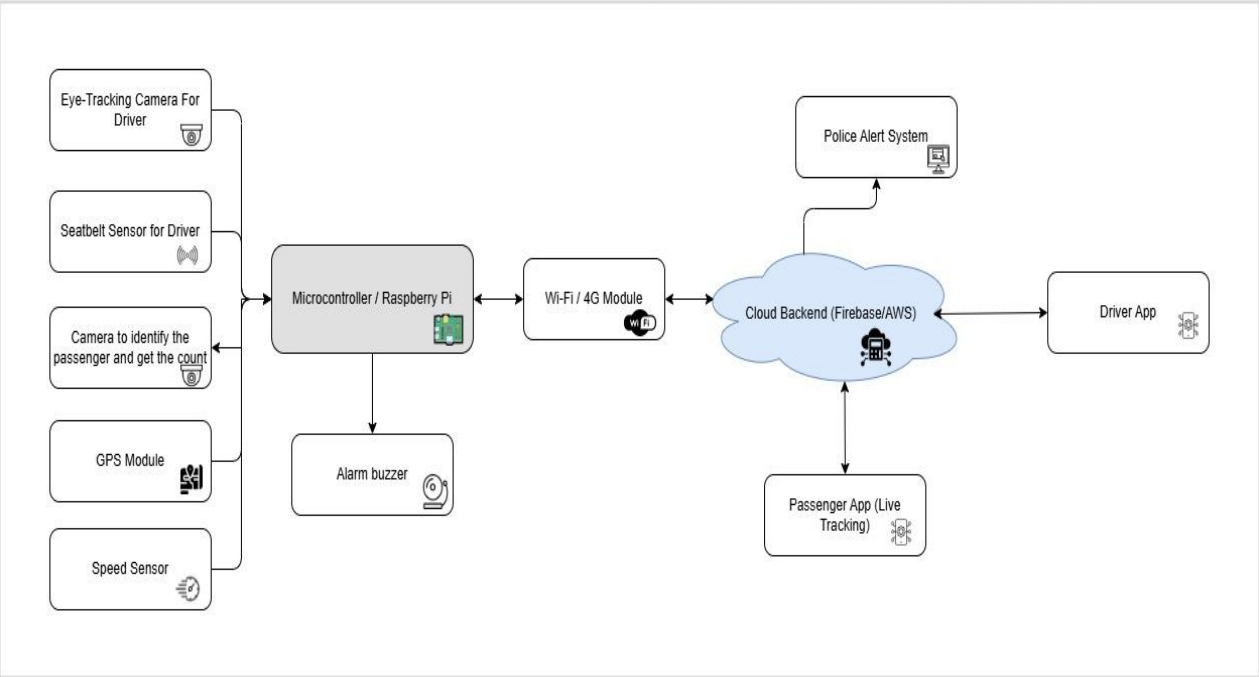
3. **Real-Time Passenger Monitoring and ID-Based Entry System: -** The Real-Time Occupancy Level Indicator monitors the number of passengers by comparing entry and exit data. It visually displays load status using a color-coded system. When the bus reaches full capacity, the system automatically plays a multilingual voice prompt to inform passengers. In parallel, the system integrates an ID-Based Identification mechanism using NFC or QR cards, assigned to each passenger—especially students—to uniquely identify and log their entry during boarding This dual-function system enhances both safety and transparency in public transportation.

4. **Live Bus Tracking**: This system enables real-time monitoring of drivers to ensure compliance with traffic rules. It records the number of violations committed by a driver within a specific distance and provides immediate feedback. If the driver accumulates three violations, the system will take appropriate action, such as issuing a warning, notifying authorities, or initiating a penalty process.

All modules are managed by a central embedded controller. Communication between components and the backend server is handled using protocols such as MQTT and REST APIs. The backend system (hosted on Firebase or AWS) processes sensor data, triggers alerts, and displays analytics through a user-friendly dashboard.

Conceptual Diagram :

A central bus unit connects sensors (eye-tracker, GPS, IR) to a microcontroller, which relays data to a cloud backend. The backend supports alert generation, a dashboard interface, and a public passenger app.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research requires interdisciplinary expertise in **embedded systems, AI-based image processing, IoT architecture, and real-time cloud communication**. Team members will work under the technical guidance of the **Centre of Excellence for AI** to ensure a high level of system integration and algorithmic optimization.

Key technical competencies include:

- **Computer Vision**: Implementation of drowsiness detection using OpenCV or Dlib for eye tracking.
- **Embedded Systems**: Programming and integration of ESP32/Raspberry Pi/Arduino platforms with peripheral sensors such as AI cameras,GPS, and accelerometers.
- **Communication Protocols**: Familiarity with MQTT and REST APIs to facilitate reliable, low-latency data transmission.
- **Cloud Development**: Real-time data processing, alerting, and visualization using Firebase, Node.js, and related technologies.
- **Mobile/Web Development**: Building intuitive passenger-facing apps for route tracking and occupancy updates.

Data Requirements:

- **Sensor Data**: Real-time data from GPS, eye-tracking, and speed sensors for event detection.
- **Occupancy Patterns**: Passenger entry/exit counts to determine peak congestion periods.
- **Traffic Violation Criteria**: Speed limits, route maps, and stop rules for compliance validation.
- **Operational Testing**: Requires access to live bus environments or simulators to capture and evaluate real-world scenarios.

8. Objectives and Novelty

Main Objective To design and implement an integrated IoT-based real-time safety monitoring system for public transport in Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
I.H.J Premathilaka IT22578532	Detect driver drowsiness and seatbelt usage	• Real time monitor driver drowsiness using AI camera • Monitor driver seat belt • Notify the driver with an alarm and voice message if he falls asleep or is tired.	This system uniquely combines real-time drowsiness detection using AI-based eye tracking with seatbelt status monitoring in a single embedded device. It provides immediate voice alerts without relying on cloud systems, making it ideal for low-bandwidth public transport. By merging biometric monitoring and physical safety checks, it offers a proactive, low-cost safety solution tailored for Sri Lankan bus drivers.

A.M.S.S Abeykoon It22592156	Identifying and reporting unsafe driving patterns	• Monitoring the driver's driving pattern and continuous driving time in real time. • The driver's driving pattern is analyzed by an AI and if it is unsafe, the driver is warned it • If the driver continues to drive recklessly, the nearest traffic police officer will be notified through our app. notify the nearest traffic police officer through our app.	Data taken from accelerometer, gyroscope and GPS input is analyzed using AI to determine whether it is a safe driving pattern or an unsafe driving pattern.
M.A.S.K Marashinghe IT22579768	Monitor and control passenger overloading	• **Real-Time Passenger Level Indicator with Voice Prompt: -** Monitor and display the current number of passengers on the bus by comparing entry and exit counts. Display real-time load status (e.g., Green = Available, Yellow = Near Full, Red = Overload). When the bus is overloaded, the system auto-plays a voice prompt: *"Attention: The bus has reached full capacity. Please wait for the next one."*. Supports multi-language voice alerts: English, Sinhala, Tamil.	In the Passenger Counting function, the system automatically plays a voice prompt when the bus reaches full capacity. Since Sri Lanka is a multilingual country where Sinhala, Tamil, and English are commonly used, the voice prompt is played in **all three languages** to ensure that all passengers can clearly understand the message. Sri Lankan's parents can use the system to **track their child's location in real-time**, including **which bus the child has boarded** and **where the bus is currently located**. This feature enhances student safety by allowing

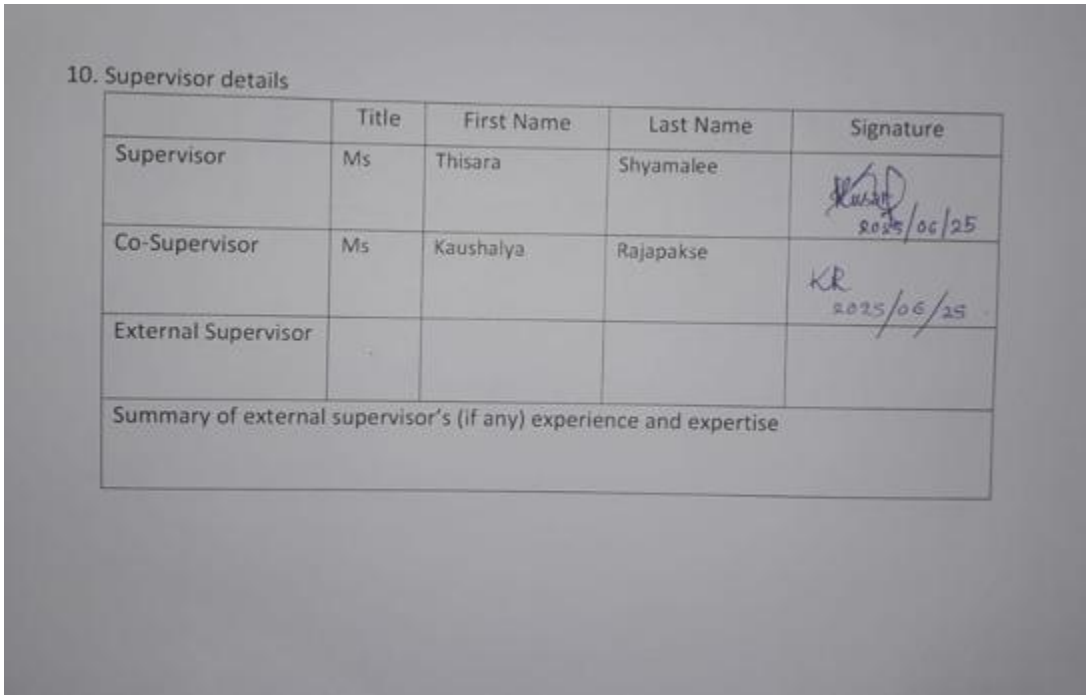
		• If the bus is unnecessarily crowded, notify the nearest police officer via notification. • **ID-Based Identification System (NFC/QR): -** Assign NFC or QR cards to passengers (especially students) to uniquely identify them and log their entry. Cards are scanned at the door during entry. This helps in tracking and auditing passenger movements. If the passenger is a student, their parents or guardians can monitor their travel activity via a connected mobile.	parents to monitor their child's journey from boarding to arrival.
S M J B Samarakoon IT22569486	Enable real-time bus tracking for commuters	• Showing the exact location of buses to passengers on a map • Passengers can register with our system • Being able to report the issues regarding the bus (To police/Transport department). • The trip of the passengers entering the bus is automatically on.	• Real-time location + occupancy displayed through public-facing app, • The passenger can view all trip details (time to destination, driver's name) • Introducing a chat section for passengers to share live update about the bus. • If any accident or delay, passengers are notified.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Premathilaka i.h.j IT22578532	**The Driver Drowsiness and Seatbelt Monitoring** component aligns with the Software Engineering specialization by integrating real-time AI-driven vision techniques and sensor-based safety validation. This component utilizes OpenCV and embedded facial landmark detection algorithms to monitor the driver's eye state, providing drowsiness alerts via a voice assistant embedded in the system. Simultaneously, a GPIO-connected physical sensor validates seatbelt usage. These features are deployed on an embedded controller (Raspberry Pi/ESP32) that supports offline functionality, ensuring immediate feedback without needing cloud connectivity. This component applies real-time feedback mechanisms, biometric computing, and autonomous alert logic to actively prevent accidents, reflecting core software engineering principles such as real-time embedded system development, sensor integration, and user-centric design.
A.M.S.S Abeykoon IT22592156	**The Unsafe Driving Pattern Detection** component is closely tied to Software Engineering practices through the design of an AI-driven analytics engine that processes multi-sensor input (accelerometer, gyroscope and GPS) to classify driving behavior. This system monitors the driver's patterns over time, identifying erratic actions like harsh braking, harsh acceleration, or continuous long-distance driving. The model triggers alerts based on safety thresholds and initiates automated notifications to authorities if the risk persists. The component involves time-series data analysis, rule-based decision automation, and edge computing—all of which represent applied knowledge in intelligent software systems and embedded decision logic.
M.A.S.K Marashinghe IT22579768	**The Passenger Overload Monitoring and ID-Based Entry System applies** software engineering knowledge in IoT-based transport systems. It leverages real-time IR/motion sensors to count passengers and identify overloads, issuing multilingual voice alerts when limits are exceeded. Furthermore, it integrates an NFC/QR-based entry system that tracks individual passengers, especially students, and logs entries with timestamps. This information is stored locally and synced with a parent-accessible app interface. The component demonstrates the application of software design for embedded systems, human-computer interaction, and secure identification, promoting both safety and transparency.

S M J B Samarakoon IT22569486	**The Real-Time Bus Tracking and Passenger Interaction Interface component** represents advanced software engineering practices through its integration of real-time GPS tracking, user data management, and interactive communication features. Passengers can view live bus locations, log their journeys, and submit feedback via a mobile/web interface. The backend integrates Firebase/Node.js for real-time updates and push notifications. It also includes a chat-based module for passengers to report delays or safety issues in real time. The system applies cloud-based software architecture, frontend-backend integration, and mobile application development tailored for public transport engagement.

10.



10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Ms	Thisara	Shyamalee	2025/06/25
Co-Supervisor	Ms	Kaushalya	Rajapakse	KR 2025/06/25
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.